KARTHIK M

Senior Databricks Data Engineer

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Professional Summary

Databricks-certified Data Engineer with 10+ years designing and building production-grade data pipelines across banking, healthcare, insurance, and telecom. Deep hands-on expertise with Apache Spark (PySpark and Spark SQL) and Delta Lake, including ACID transactions, data versioning, and multi-hop medallion architectures (Bronze/Silver/Gold). Expert-level Python and SQL applied to scalable data transformations at 5+ TB scale. Demonstrated track record optimizing data workloads for performance and cost efficiency: cut compliance reporting time 40%, accelerated pipeline runtime 35%, and reduced job execution time 30%. Proven ability to implement monitoring, observability, and error handling in regulated environments with strict security requirements (GDPR, HIPAA, banking). Strong software engineering discipline: CI/CD, test-driven development, data quality frameworks (Great Expectations), and Agile/Scrum delivery across large, complex enterprise systems.

Core Technical Skills

Area	Skills
Databricks & Spark	Azure Databricks, PySpark (Advanced), Spark SQL, Spark Streaming, Delta Lake, Delta Live Tables, Unity Catalog, Databricks Workflows
Data Architecture	Multi-hop Medallion (Bronze/Silver/Gold), Lakehouse, Lambda/Kappa patterns, ACID transactions, data versioning
Languages	Python (Expert), SQL (Expert), PL/SQL, Scala, HQL
Data Quality & Observability	Great Expectations, Matillion, SonarQube, Azure Monitor, Nagios, Prometheus
Orchestration & Integration	Apache Airflow, Azure Data Factory, Databricks Workflows, Apache Kafka, Azure Event Hubs, Azure Functions
Cloud & DevOps	Azure (Databricks, Data Lake, Synapse Analytics, Key Vault, Active Directory, Event Hub), Azure DevOps, GitHub, Jenkins, Terraform, Docker, ARM Templates

Area	Skills
Databases & Warehouses	Snowflake, Azure SQL DB, SQL Server, Oracle, Cosmos DB, MongoDB, Cassandra
BI & Reporting	Power BI, SSRS, SSAS, Tableau
Certifications	Microsoft Certified: Azure Data Engineer Associate; Power BI Data Analyst Associate

Professional Experience

Senior Databricks Data Engineer | Wells Fargo, Irving, TX

Jun 2024 – Present

- Designed and built a production-grade Databricks Lakehouse on Unity Catalog implementing a multi-hop medallion architecture (Bronze/Silver/Gold), delivering centralized governance, fine-grained access control, and full data lineage tracking across a regulated banking environment.
- Built PySpark and Delta Live Tables pipelines ingesting, cleansing, and harmonizing 5+ TB of customer complaints data, leveraging Delta Lake ACID transactions and data versioning for reliable, auditable data at the Silver and Gold layers.
- Implemented automated data quality validation using Great Expectations, enforcing schema and business-rule checks before data is promoted between medallion layers — ensuring data quality for Compliance and Risk reporting.
- Re-engineered legacy SQL into optimized Spark and DBT transformations, improving scalability, accelerating daily pipeline execution by 35%, and meaningfully reducing cluster compute costs.
- Orchestrated end-to-end data workflows across Azure Data Factory, Databricks Workflows, and Airflow, feeding curated Gold-layer data to Synapse Analytics and Power BI — cutting compliance report generation time by 40%.
- Designed standardized schemas in Hackolade for multi-vendor data feeds, enforcing consistency and preventing data quality issues before ingestion into the Delta Lake storage layer.
- Implemented CI/CD pipelines with Azure DevOps and GitHub for automated testing and zero-downtime deployments across dev, staging, and production environments.

Environment: Azure Databricks, Unity Catalog, Delta Lake, Delta Live Tables, PySpark, Spark SQL, DBT, Great Expectations, Azure Data Factory, Databricks Workflows, Airflow, Synapse Analytics, Power BI, Azure DevOps, GitHub

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Senior Data Engineer | Medtronic, Minneapolis, MN

Nov 2021 – May 2024

- Designed and built high-performance PySpark ETL pipelines on Azure Databricks for large-scale data transformations, with hands-on tuning for speed, scalability, and memory efficiency across distributed workloads.
- Built real-time data services integrating Spark Streaming and Kafka with Azure Event Hubs and Azure Functions, processing pacemaker telemetry into Snowflake for near-real-time clinical analytics — operating in a strict HIPAA-regulated environment.
- Implemented monitoring, observability, and alerting with Azure Monitor, Nagios, and Prometheus, providing full pipeline visibility and proactive error detection across distributed systems.
- Applied data governance and automated quality controls, combined with Azure Policy, to enforce GDPR/HIPAA compliance across all data subscriptions and resource groups.
- Hardened pipeline security with encryption, firewalls, intrusion detection, and role-based access controls; conducted regular audits and vulnerability assessments to meet strict security requirements.
- Integrated SonarQube into CI/CD pipelines to enforce code quality standards and software engineering best practices on all data pipeline code.
- Automated infrastructure provisioning with ARM templates and Azure CLI, and integrated Snowflake with Power BI and Azure Analysis Services for self-service dashboards.

Environment: Azure Databricks, PySpark, Spark Streaming, Kafka, Snowflake, Azure Event Hubs, Azure Functions, Azure Data Lake, Azure Policy, Azure Monitor, Nagios, Prometheus, SonarQube, ARM Templates, Power BI, Microsoft Fabric

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Data Engineer | State Farm Insurance, Richardson, TX

Apr 2020 – Oct 2021

- Designed and delivered end-to-end production data pipelines integrating DB2, SQL, Oracle, flat files, APIs, XML, and JSON via Azure Linked Services into a scalable Databricks processing layer.
- Implemented Spark Streaming on Azure Databricks and Azure EventHub for real-time processing of high-volume insurance data feeds.
- Optimized Apache Spark workloads for performance and cost efficiency — tuning resource allocation, partitioning, and shuffle configuration — reducing job execution time by 30%.
- Built Terraform modules to provision Azure infrastructure (VNets, VMs, AKS clusters) and managed Docker-based portable pipeline environments, enforcing infrastructure-as-code discipline.
- Set up Azure DevOps YAML pipelines for CI/CD of data pipelines across environments, and used Matillion to enforce data quality and streamline ETL integration.

Environment: Azure Databricks, PySpark, Spark Streaming, Spark SQL, Azure EventHub, Terraform, Docker, Azure Synapse Analytics, Matillion, Tableau, Azure DevOps

Data Engineer | Centene Corporation, Dallas, TX

Jul 2018 – Mar 2020

- Built Spark and PySpark batch transformation pipelines in Python and Scala to process large unstructured healthcare datasets, and developed reusable Azure Data Factory pipelines for data-driven workflows.
- Created Databricks ETL pipelines using notebooks, Spark DataFrames, and Spark SQL, automating ingestion workflows and database updates at scale.
- Implemented Informatica Data Governance (Data Privacy Management, Enterprise Data Catalog) and BigID for automated data discovery and privacy compliance.
- Worked in Agile/Scrum with test-driven development, running unit and integration tests on all delivered pipelines — ensuring correctness and ownership over every deliverable.

Environment: Spark, PySpark, Python, Scala, Azure Data Factory, Databricks, Spark SQL, Informatica PowerExchange, PolyBase, Azure SQL Data Warehouse, Power BI, BigID

ETL Developer | Nielsen Corporation, Tampa, FL

Oct 2016 – Jun 2018

- Delivered production ETL and reporting solutions on SQL Server; developed and migrated DTS packages to SSIS with zero data loss.
- Tuned stored procedures, queries, and database objects (tables, views, functions) to improve performance in large-scale data warehouse environments.
- Designed SSAS cubes using Star Schema with MDX for KPIs and cell-level security; built Power BI and SSRS reports on top of SSAS Tabular models.

Environment: MSBI, SSIS, SQL Server, SSAS, MDX, SSRS, Power BI, Star Schema

Data Warehouse Developer | WellCare, Wilmington, NC

Jan 2015 – Sep 2016

- Built Informatica ETL mappings extracting and transforming data from Oracle, Sybase, SAP, flat files, XML, and SQL Server.
- Implemented Slowly Changing Dimension (SCD) Type 2 logic for historical data accuracy in regulated healthcare reporting.

Environment: Informatica PowerCenter, SAP HANA Studio, Oracle, SQL Server, SCD Type 2, Unix/Linux

Data Warehouse Developer | Truist Bank, Atlanta, GA

Jan 2014 – Dec 2014

- Developed complex stored procedures, triggers, and functions; designed ETL data flows in SSIS migrating and transforming data from multiple sources.
- Built SSAS cubes with aggregations, KPIs, and data mining models; developed parameterized SSRS reports for business stakeholders.

Environment: SQL Server, SSIS, SSAS, MDX, SSRS, Dimensional Modeling, SCD

Education

M.S., Computer Science — Stevens Institute of Technology | *December 2013*

B.S., Computer Science and Engineering — Osmania University | *May 2012*